ABHIJEET SAHDEV

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SUMMARY

Machine Learning Engineer with expertise in full-stack development, cloud infrastructure, and large-scale dataset engineering. Experienced in building and deploying end-to-end ML systems that bridge research and production, delivering measurable improvements in accuracy, efficiency, and scalability.

EXPERIENCE

Research Assistant

Immersive Creation Lab [Code]

May 2025 – Ongoing | Remote

- Developed a web-based 3D Copilot, an innovative research prototype for ACM UIST 2026, enabling natural language-driven 3D scene manipulation using React and Three.js, with a focus on AI-assisted user interfaces for intuitive 3D modeling.
- Designed an interactive UI with a shape palette, object manipulator, and real-time 3D canvas featuring raycasting, transform controls, and OrbitControls, optimizing usability and supporting precise object manipulation while addressing edge cases like ambiguous commands and resource cleanup.
- Implemented Sketchfab API integration for seamless 3D model retrieval, incorporating ZIP extraction with JSZip for GLTF/GLB handling, model normalization, and efficient blob URL management to enhance scene integration and rendering.
- Integrated OpenAI's GPT-4o-mini via a Node.js/Express backend to parse complex user commands, supporting actions like adding shapes, manipulating objects, searching Sketchfab, and managing scenes with multi-command processing.

Graduate and Undergraduate Teaching Assistant

NJIT

Aug 2024 – Dec 2024 | Newark

- Responsible for grading various phases of capstone project and setting up evaluation metrics for assignments of CS Data Mining for 20+ students.
- Invigilated mid-term exams for DS Machine Learning course for 70+ students.

Software Engineering Intern

MyHealthToday [Report]

Dec 2020 – Jun 2021 | Remote

- Developed a cross-platform front-end using React for web browsers and React Native for mobile apps, ensuring responsiveness across devices through collaboration with the CTO and technical problem solving.
- Created comprehensive test suites to verify application adherence to design specifications and compatibility with target customer devices, enhancing reliability.
- Implemented design patterns (adapter, state, observer) using React-Redux while typing maintainable code to establish a single source of truth, optimizing state management for the mobile app.
- Standardized record formats in DynamoDB tables via scripting, proactively preventing future run-time errors.
- Developed microservices using serverless AWS Lambda in a distributed, event-driven architecture, contributing to a scalable and fault-tolerant backend serving both mobile and Alexa applications.
- Designed, implemented, and deployed RESTful APIs using SAM CLI and AWS CI/CD tools, streamlining resource creation and reducing client-side processing time by 16%.

SKILLS

- Python, Javascript, C++, Java, Dart
- scikit-learn, PyTorch, TensorFlow
- Django, React, React Native, Flutter
- Git, AWS, GCP, Docker
- MySQL, DynamoDB

EDUCATION

MS in Artificial Intelligence

New Jersey Institute Of Technology | CGPA: 3.86/4

Jan 2024 - Dec 2025 | Newark, NJ

B.Tech in Computer Science and Engineering, Minor in Computational Mathematics

MIT, Manipal Academy of Higher Education

Jun 2017 – Aug 2021 | Udupi, India

PUBLICATIONS

Statistical Analysis of Sentence Structures through ASCII, Lexical Alignment and PCA

A. SAHDEV [Report][Code]

Mar 2025

PROJECTS

GIT-base Ablation Study

PyTorch [Report][Code]

Aug 2025

- Engineered and preprocessed MIMIC-CXR-JPG v2.1.0 (377K+ chest X-rays, 227K reports) via GCP BigQuery and CheXpert, generating structured mini-reports and building a balanced 66K+ study dataset with robust stratified splits
- Implemented and benchmarked GIT-base vision–language architecture under Full Fine-Tuning vs. LoRA (vision-only, text-only), designing a resource-optimized training strategy that reduced GPU memory usage by 50% and training time by 67%.
- Demonstrated 5.8× improvement in BLEU with LoRA Text-Only over full fine-tuning, validating LoRA as a parameter-efficient and domain-adaptive strategy for medical vision–language tasks.

Study of Selected Clustering Algorithms with Generated Datasets

Python, Scikit-Learn [Report] [Code]

Nov 2024

- Performed data mining in benchmarking unsupervised ML algorithms such as K-Means and Hierarchical Agglomerative Clustering on AI-generated datasets with outliers removed using IQR to enhance pattern discovery.
- Achieved top silhouette scores with K-Means on Dataset 1 (0.359, 3 clusters) and Dataset 2 (0.625, 2 clusters), and with Agglomerative on Dataset 3 (0.984, 2 clusters).

Open Source Contribution

React Native, JS [Pull Request]

Mar 2021

- Introduced a boolean argument to display number of selected_items in react-native-multiple-select by toystars **[550+ stars, 500k+ downloads]**.